

Lesson 3-5 · Period 3

Real + Complex Zeros, and Modeling

Klmsara-adapted · Student-centered · No Blooket

DOK 1–2

5 min

bridge from Tuesday

Practice (3-5-savvas-q11) · Where do the graphs cross? At what points do the graphs of $f(x) = x^3 - 2x^2 - 16x + 20$ and $g(x) = -12$ intersect?

Graph is on your packet. List all intersection points.

Bridge (verbal): A cubic has zeros 2 , $1 + \sqrt{3}$, and $1 - \sqrt{3}$. Are the coefficients rational, irrational, or complex? Why?

DOK 2

12 min

teacher models

Complex-Zero Rules (keep on your page)

1. Complex zeros of a real-coefficient poly come in **CONJUGATE PAIRS**.
2. If $a + bi$ is a zero, then $a - bi$ must also be a zero.
3. A degree- n polynomial has exactly n zeros (counting multiplicity).
4. Example: cubic with real zero $x = 4$ and complex zero $2 + 3i \Rightarrow$ third zero is $2 - 3i$.

Modeling Rules (Storage Box)

1. Write $V(x)$ in factored form; each factor = a physical dimension.
2. Use the constraint ("length is greatest") to label length, width, height.

DOK 2–3

38 min

student work

4 items in order. Plan ~20 min for Practice #27 (Storage Box).

Try It 3a · Real + complex zeros (cubic) $f(x) = 2x^3 - 8x^2 + 9x - 9$. Find ALL zeros.

Try It 3b · Real + complex zeros (quartic) $f(x) = x^4 - 3x^2 - 4$. Find ALL zeros.

Practice #17 · Real + complex zeros $f(x) = x^3 - x^2 - 4x - 6$. Find ALL zeros.

DOK 2

7 min

DOK-3 reveal + assessment preview

Sentence frame · Storage Box “For the Storage Box, the factored form was _____. I chose _____ as the length because _____. To find $V = 10 \text{ ft}^3$, I _____.”

Connection to Thursday's Assessment Assessment Q#6 (Lucy's tray) uses the **same move**: factored volume model $\rightarrow$ extract a hidden dimension. Today's Storage Box is your rehearsal.

DOK 1–2

3 min

not graded

Complete on your exit ticket

1. Today I learned that _____.
2. For the Storage Box, the hardest step was _____.
3. Thursday's assessment will test _____.